Mohammad Uzair Khan

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Education

BITS Pilani, Dubai Campus — Dubai, UAE
B.E. Computer Science — *Expected 2027*
- Relevant Coursework: Database Management Systems, Machine Learning, Data Structures & Algorithms, Large Language Models

Bishop Conrad Senior Secondary School | *Senior Secondary (Class XII)* — Bareilly, India | *2023*

Profile

Computer Science undergraduate doing **pharmaceutical commercial analytics**: **data transformation and harmonization** of multi-million-row public US healthcare data, **data analysis** in Python and SQL, and **Excel** and **Power BI** reports that turn it into **actionable insights**.

Data Analytics Projects

Rx Market Pulse: Medicaid drug spend, 2019–2025 | *Python, SQL (DuckDB), Excel, Power BI* — Live | Code
- Built a tested pipeline over **36.1M** public Medicaid rows; **harmonized 1,488** manufacturer name variants with **19** data-quality checks; a **Power BI** report (star schema, 28 DAX measures, 6 pages) and a web dashboard share one set of numbers
- Insight: spend rose **60% to $116.9B** on flat prescriptions; brands are **14%** of prescriptions and **89%** of spend; live-formula **Excel** territory model ranks 51 states, checked in Excel through 6 scenarios against Python

Payer Access Intelligence: BELSOMRA vs peers in Medicare | *Python, SQL, Excel, Power BI* — Report | Code
- Harmonized **2.24M** CMS formulary records across **367** Medicare drug plans and eight strengths; joined two 2026 snapshots to enrollment without double counting; **20** audit checks; Excel and Power BI (29 DAX measures)
- BELSOMRA is listed in plans holding **80.5%** of members (DAYVIGO 17.7%, QUVIVIQ 9.3%); its rise from 79.5% was enrollment mix, not new listings; **70 plans (3.2M members)** queued for payer review

Field Force Planner: sales-force sizing and territories | *Python, SQL (DuckDB), Excel* — Live | Code
- Transformed the **9.9M**-record NPI registry, placed **98.9%** of providers in a county and sized a launch at **1,310 diabetes** and **474 oncology** reps; Excel sizing model checked in Excel on **2,730 cells**
- Aligned 3,143 counties into connected territories: workload spread **7.5%** vs 274% by county count; **99.6%** of territories within 25% of target

Drug Price Watch: US pharmacy drug prices | *Python, SQL, Excel, GitHub Actions* — Live | Code
- Built a price index over **17.3M** weekly NADAC prices (every year ties to the portal): since 2019 brand prices **+22%**, generics **–53%**; **53%** of brand price steps fall in January and July
- Automated weekly reporting with GitHub Actions: it found **28 generic sitagliptin packs** at **36%** of Januvia's price; cross-checked Rx Market Pulse's price effect

Trial Pipeline Benchmark: MSD vs 11 peers | *Python, REST API (JSON), SQL, Excel* — Live | Code
- Pulled **36,612** trials from the ClinicalTrials.gov API and harmonized **1,304** sponsor names to 12 companies (**25** checks, **264** tests); MSD has the **second-largest** active pipeline (238 studies), **35%** of Phase 2+ on Keytruda

Also built: CampaignOps *(campaign KPIs, synthetic data; code)* · PharmaEvidence · SignalDesk AI — GitHub

Experience

Backend Developer Intern, Amaani (Travel-Tech Startup) — Dubai, UAE
BITS Practice School (PS-1) — *Jun 2025 – Aug 2025*
- Developed and maintained backend services in **Python (Flask)**, building RESTful APIs for core travel-booking features
- Documented API contracts with **Swagger/OpenAPI** and validated endpoints end-to-end using **Postman**; implemented request validation and data models

Research & Publications

Lightweight Deep Learning for Tomato Leaf Disease Detection on Edge Devices — 2025 – 2026
First Author; submitted to IEEE MSN 2026 (under review)
- Trained a MobileNetV2 classifier (10 classes) reaching **97.33% test accuracy** (macro-F1 0.963); compressed it to **5.3 MB** with INT8 quantization and ran it on a Raspberry Pi 4B at about 250 ms per image

Technical Skills & Certification

Data Analysis & Insights: data analysis, commercial analytics, KPI and variance analysis, price indexes, forecasting, problem solving
Excel: SUMIFS, INDEX/MATCH, MAXIFS, PERCENTRANK, live-formula and scenario models, charts
Visualization: Power BI (star schema, DAX), Excel charts, interactive web dashboards (Next.js/React)
Transformation & Harmonization: Python (Pandas, NumPy), SQL (DuckDB, MySQL), data cleaning, data-quality checks, GitHub Actions
Certification: The Ultimate Job Ready Data Science Course, CodeWithHarry (2026)